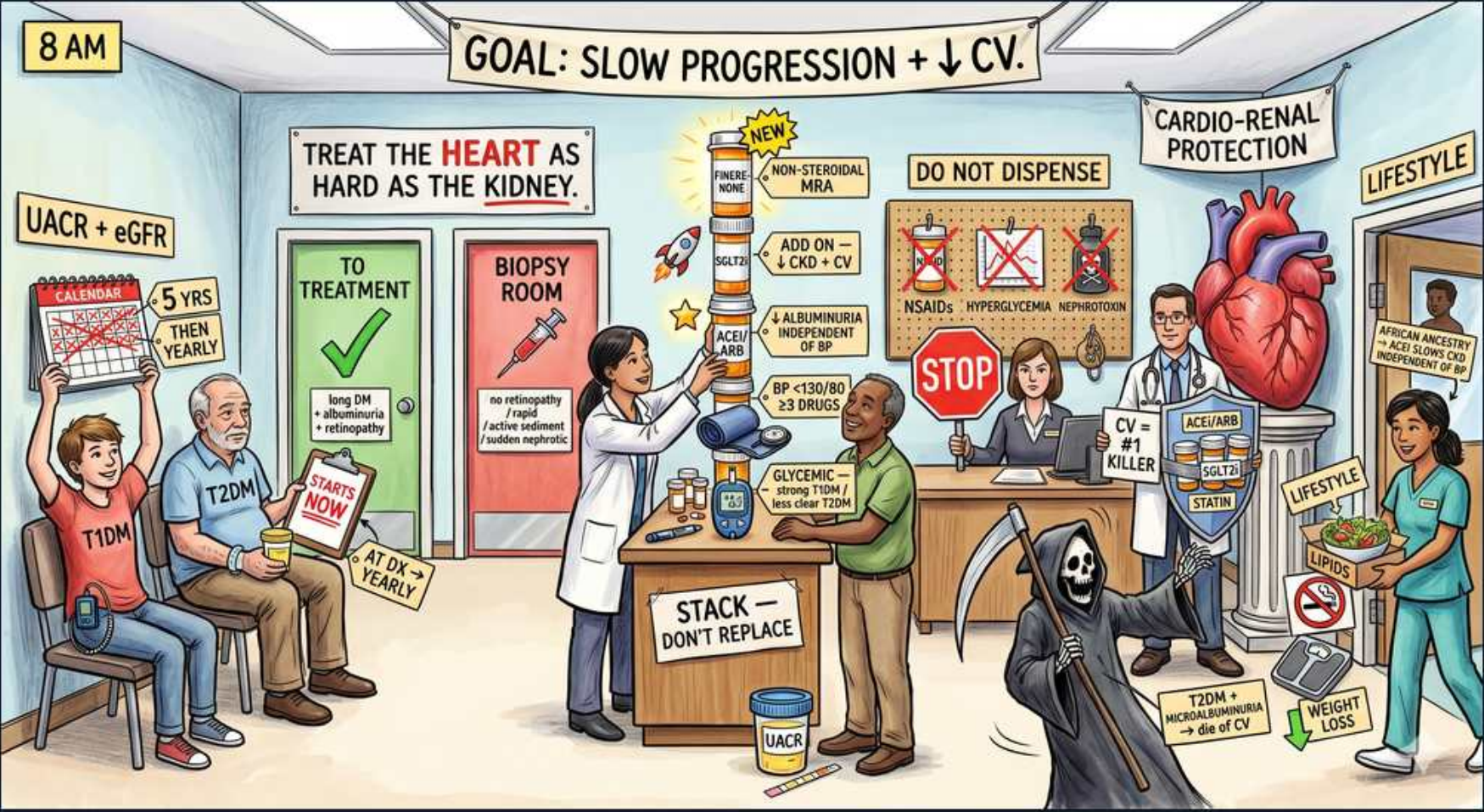


Diabetic Nephropathy — Management & Cardio-Renal Protection



1. Goal: slow progression + ↓CV

WHAT YOU SEE

Banner: goal is slow progression + reduce cardiovascular risk

WHAT IT ENCODES

The overarching aims in diabetic kidney disease are to slow GFR decline/albuminuria progression AND reduce cardiovascular events, the leading cause of death. Therapy is multifactorial: glycemic, BP, RAAS blockade, SGLT2i, and lifestyle.

BOARD TRAP

Renal endpoints alone are insufficient — most patients die of cardiovascular disease, so cardio-renal protection is the real goal.

2. Screen UACR + eGFR yearly

WHAT YOU SEE

Calendar: UACR + eGFR yearly; T1DM at 5 yrs, T2DM at diagnosis

WHAT IT ENCODES

Screen with annual urine albumin-to-creatinine ratio AND eGFR. Begin screening 5 years after diagnosis in type 1 DM and AT diagnosis in type 2 DM (since onset is unknown). Confirm albuminuria on 2 of 3 samples.

BOARD TRAP

T1DM screening starts at 5 years; T2DM starts AT diagnosis — because type 2 has been present subclinically for years.

3. Biopsy red flags

WHAT YOU SEE

Biopsy room: red flags = no retinopathy, rapid/active sediment, sudden nephrotic

WHAT IT ENCODES

Biopsy is reserved for atypical features suggesting non-diabetic disease: absence of retinopathy, rapidly rising creatinine, active urinary sediment (RBC casts), or sudden nephrotic-range proteinuria. Typical diabetic nephropathy needs no biopsy.

BOARD TRAP

Long-duration diabetes + retinopathy + slow proteinuria = clinical diagnosis, no biopsy; ABSENT retinopathy or active sediment should trigger a biopsy for alternative GN.

4. Stack — don't replace (ACE/ARB base)

WHAT YOU SEE

Medication stack: ACE/ARB base, stack don't replace

WHAT IT ENCODES

Therapy is additive ('stack, don't replace'). The foundation is an ACE inhibitor OR ARB (never both together) for any albuminuria, titrated to the maximum tolerated dose, on top of which SGLT2i, GLP-1, and nonsteroidal MRA are added.

BOARD TRAP

Never combine an ACE inhibitor WITH an ARB — dual RAAS blockade increases hyperkalemia and AKI without added benefit.

5. SGLT2 inhibitor

WHAT YOU SEE

Rocket/tower: SGLT2i, renoprotective independent of glucose, add on if CKD+CV

WHAT IT ENCODES

SGLT2 inhibitors reduce albuminuria, slow GFR decline, and cut CV/heart-failure events — benefits largely INDEPENDENT of glucose lowering, via restored tubuloglomerular feedback (afferent constriction). Add for diabetic CKD with eGFR above threshold.

BOARD TRAP

SGLT2i renoprotection is independent of glycemic control — give them for CKD/CV benefit even when A1c is at goal.

6. Nonsteroidal MRA (finerenone)

WHAT YOU SEE

Sign: non-steroidal MRA, add on if CKD + CV

WHAT IT ENCODES

Nonsteroidal mineralocorticoid receptor antagonists (finerenone) further reduce albuminuria and CKD progression/CV events when added to a RAAS blocker in diabetic CKD, with less hyperkalemia than spironolactone.

BOARD TRAP

Finerenone (nonsteroidal MRA) is added ON TOP of an ACE/ARB — watch potassium; it is not a replacement for RAAS blockade.

7. Glycemic — strong T1DM, less in CKD

WHAT YOU SEE

Glycemic control: strong in T1DM, less benefit in advanced CKD

WHAT IT ENCODES

Tight glycemic control (target A1c ~7%) clearly slows nephropathy onset/progression early (esp. T1DM), but benefit attenuates in advanced CKD where hypoglycemia risk rises. Individualize targets in late-stage disease.

BOARD TRAP

In ADVANCED CKD, ultra-tight glycemic control yields little renal benefit and raises hypoglycemia risk — loosen the A1c target.

8. BP <130/80

WHAT YOU SEE

BP target <130/80

WHAT IT ENCODES

Blood pressure control (generally <130/80) is central; ACE inhibitors/ARBs are preferred agents because they reduce intraglomerular pressure and proteinuria beyond their BP effect.

BOARD TRAP

Prefer a RAAS-blocking agent for hypertension in diabetic nephropathy — equal BP lowering with a non-RAAS drug gives less renal protection.

9. Treat heart as hard as kidney

WHAT YOU SEE

Sign: treat the heart as hard as the kidney

WHAT IT ENCODES

Cardiovascular disease is the #1 killer in diabetic nephropathy; aggressive lipid lowering (statins), antiplatelet where indicated, and the cardio-protective SGLT2i/GLP-1 agents are integral. Cardio-renal management is unified.

BOARD TRAP

Don't focus only on the kidney — CV death outpaces ESKD; statins and CV risk reduction are mandatory.

10. NSAIDs — do not dispense

WHAT YOU SEE

Pharmacist STOP: NSAIDs, hyperglycemia, nephrotoxins to avoid

WHAT IT ENCODES

Avoid nephrotoxins: NSAIDs (reduce GFR, worsen hyperkalemia with RAAS blockade), nephrotoxic contrast where avoidable, and uncontrolled hyperglycemia. Dose-adjust renally cleared drugs.

BOARD TRAP

NSAIDs blunt afferent vasodilation and stack with ACE/ARB to cause AKI and hyperkalemia — counsel patients to avoid them.

11. GLP-1 receptor agonist

WHAT YOU SEE

New: GLP-1 RA, cardio-renal benefit

WHAT IT ENCODES

GLP-1 receptor agonists reduce albuminuria and provide cardiovascular benefit (and weight loss) in diabetic kidney disease, complementing SGLT2 inhibitors as part of modern cardio-renal therapy.

BOARD TRAP

GLP-1 RAs add CV/renal and weight benefit on top of SGLT2i — they are complementary, not interchangeable.

12. Lifestyle + weight loss

WHAT YOU SEE

Lifestyle corner: diet, weight loss, smoking cessation

WHAT IT ENCODES

Lifestyle measures — sodium restriction, weight loss, smoking cessation, and exercise — reinforce pharmacologic therapy in slowing progression and cutting CV risk.

BOARD TRAP

Lifestyle change is foundational and additive, never a substitute for RAAS blockade and SGLT2i in established diabetic nephropathy.

13. T2DM + microalbuminuria = ↑CV death

WHAT YOU SEE

Heart shield: CV is #1 killer; microalbuminuria marks CV death

WHAT IT ENCODES

Even moderate albuminuria is an independent marker of cardiovascular mortality in type 2 diabetes — it flags global vascular risk, not just renal risk, and should intensify CV prevention.

BOARD TRAP

Microalbuminuria is not 'just kidney' — it independently predicts cardiovascular death and warrants aggressive CV risk reduction.
